

EEE 359

Engineering Project Management

Lecture 09

Quality Management

Quality Management

Project Quality Management includes the processes for incorporating the organization's quality policy regarding **planning, managing, and controlling** project and product quality requirements in order to meet stakeholders' objectives.

Project Quality Management also supports **continuous process improvement** activities as undertaken on behalf of the performing organization.

Quality and **grade** are not the same concepts. Quality refers to how well a product meets specifications and fulfills its intended purpose (fitness for use), whereas grade is the category or rank assigned to products with similar functions but different technical characteristics

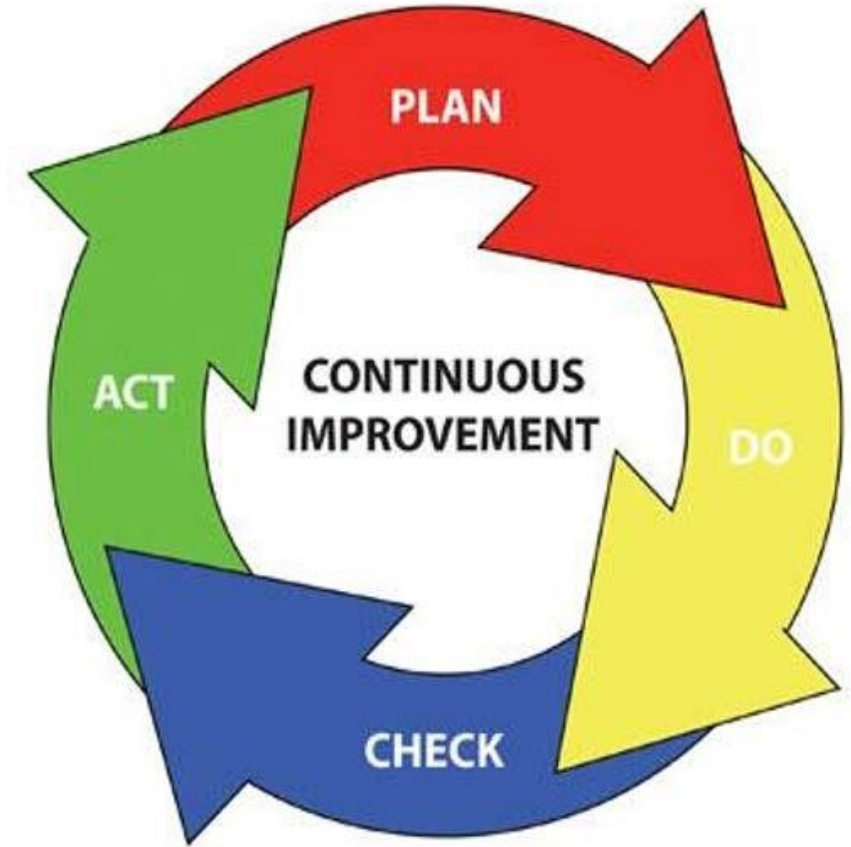
A low-grade product (e.g., base model car) can be high-quality, while a high-grade product (e.g., luxury car) can be low-quality if it is unreliable.

Continuous Improvement

- An ongoing effort to enhance products, services, or processes through **small, incremental changes** rather than sudden, radical shifts.
- In business, it refers to **continuously improving** quality in all areas. The entire company looks for and initiates small improvements in quality throughout.
- Originated from Japanese philosophy of "**Kaizen**", meaning "change for the better".
- Common Methodologies:
 - **Lean Manufacturing** - minimizing waste while maximizing productivity and value for the customer.
 - **Six Sigma** - reducing variation, eliminating defects.

Continuous Improvement - the PDCA cycle

- **Plan-** Recognize an opportunity and plan a change.
- **Do-** Test the change. Carry out a small-scale study.
- **Check-** Use what you learned to plan new improvements, beginning the cycle again.
- **Act-** Take action based on what you learned. If you were successful, incorporate what you learned from the test into wider changes.



Control Charts

- Used to determine if a process is stable and predictable.
- Used to monitor processes to ensure that resulting production “piece- parts” are within acceptable limits, or to determine if any actions are required.
- Can also be used to monitor project performance, e.g. cost and schedule variances.
- Usually measured by Quality Control.
- Production items outside a given specification limit are defective.

Control Charts

- **Mean (μ)**
 - The mean is indicated by a line in the middle of the control chart.
- **Specification Limits** represent the customer's standards or contractual requirements for performance and quality.
- **Upper and Lower Control Limits**
 - The acceptable range of variation of a stable process. **Every process has some variation.** Data points inside the range are usually considered “in control.”
 - The **acceptable range** between the upper and lower control limits is statistically calculated by the project manager and stakeholders.
 - **Usually at +/- 3σ (sigma) from the mean**
 - Often shown as two dashed lines

Control Charts

Out of Control: Indicates a lack of process stability, consistency and predictability due to a **potential assignable cause**.

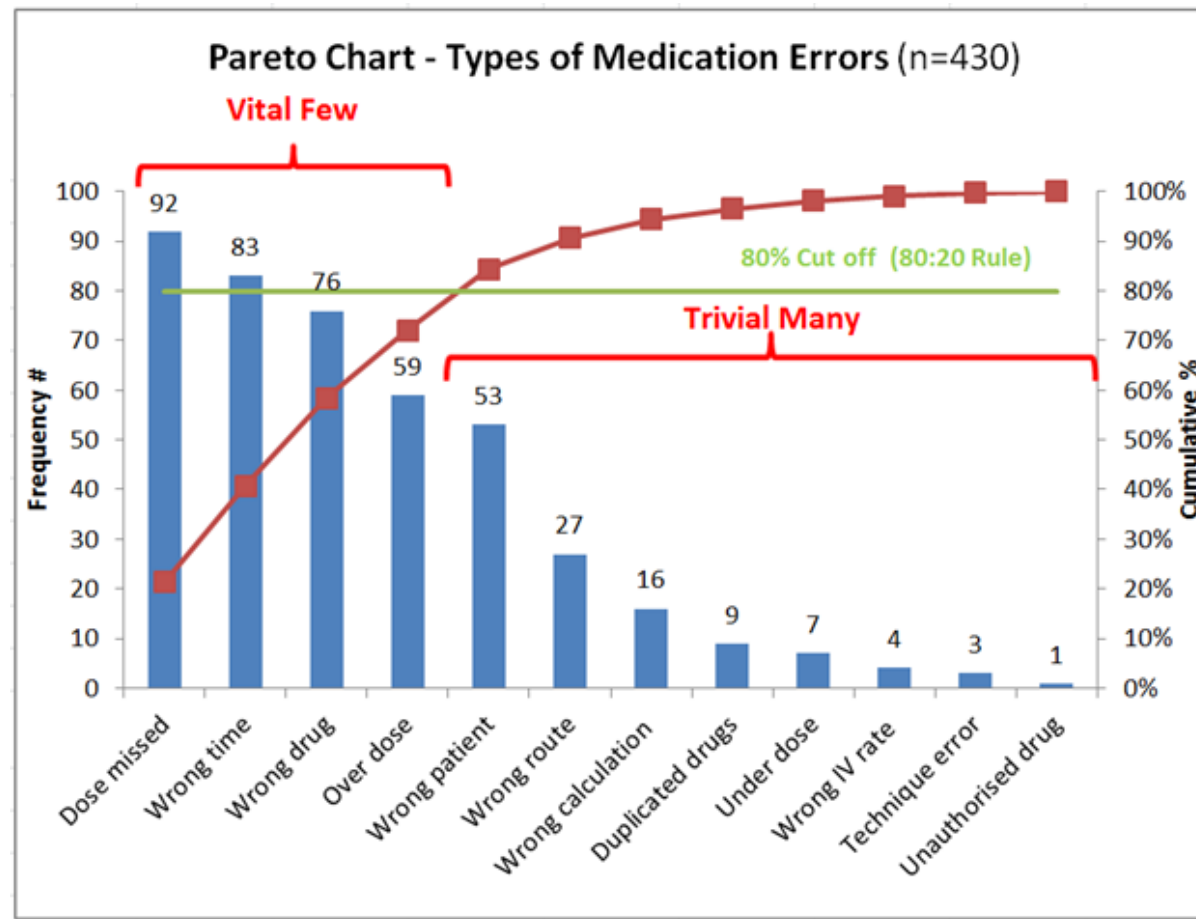
- Two (2) circumstances where the process is “out of control”
 1. A data point falls outside the upper or lower control limit
 2. The Rule of Seven - a run of seven or more consecutive points in the control chart, either above the mean, or below the mean, or continuously increasing or decreasing
- The probability of occurrence is so small (0.0078), the process may be out of control even though none of the data points are outside the control limits.

Control Charts



Pareto Chart

Illustrates the number of quality defects generated by which causes. Example: Shows six (6) common causes of poor quality wine. Chart displays the number of times it has generated poor quality wine.



80/20 Rule:

- 80% of all problems are due to 20% of causes.
- Helps to identify the most critical issues by ranking them left to right.

Thanks